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Math 201 Final Exam Term 243

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The slope of the parametric curve $x = 2(\theta - \sin\theta)$, $y = 2(1 - \cos\theta)$ at the point corresponding to $\theta = \dfrac{\pi}{6}$ is equal to

(a) $2 - \sqrt{3}$

(b) 10

(c) $\sqrt{3}$

(d) $1 + \sqrt{3}$

(e) $2 + \sqrt{3}$

2. Let $f(x, y) = x^2 + xy$, and $R = \{(x, y) : |x| \leq 2,\ |y| \leq 1\}$. If M and m represent respectively the absolute maximum and absolute minimum of f over R, then $m + M =$

(a) 6

(b) $\dfrac{21}{4}$

(c) $\dfrac{25}{4}$

(d) $\dfrac{11}{2}$

(e) $\dfrac{23}{4}$

3. The volume of the solid inside the sphere $x^2 + y^2 + z^2 = 4$, outside the cone $z = \sqrt{x^2 + y^2}$, and above the xy-plane is

(a) $8\sqrt{2}\,\pi$

(b) $\dfrac{8\pi}{3}$

(c) $\dfrac{16\sqrt{2}\,\pi}{3}$

(d) $\dfrac{8\sqrt{2}\,\pi}{3}$

(e) $\dfrac{4\sqrt{2}\,\pi}{3}$

4. The maximum value of the directional derivative of $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ at the point $P(1, 4, 2)$ is

(a) 2

(b) $\dfrac{1}{\sqrt{21}}$

(c) 1

(d) 0

(e) $\sqrt{21}$

5. Let $P(a, b, c)$, with $a > 0$, be the point on the ellipsoid $x^2 + 4y^2 + z^2 = 9$ where the tangent plane is perpendicular to the line

$$L: \ x = 2 - 4t, \quad y = 1 + 8t, \quad z = 3 - 2t.$$

Then $a + b + c =$

(a) 5

(b) 1

(c) 2

(d) 6

(e) 0

6. A rectangular box is resting on the xy-plane with one vertex at the origin and its opposite vertex lying in the plane $5x + 3y + 2z = 30$. The maximum possible volume of the box is equal to

(a) $\dfrac{100}{3}$

(b) $\dfrac{200}{3}$

(c) 90

(d) 50

(e) $\dfrac{125}{3}$

7. The minimum value of $f(x, y) = \dfrac{3}{2}x + y + 1$ subject to the constraint $x^2y = 6$ is equal to

(a) 14

(b) $\dfrac{17}{2}$

(c) 9

(d) $\dfrac{15}{2}$

(e) $\dfrac{11}{2}$

8. The average value of $f(x, y) = \sin(x + y)$ over the rectangular region with vertices $(0, 0)$, $(\pi, 0)$, (π, π) and $(0, \pi)$ is equal to

(a) 0

(b) 2π

(c) π

(d) 3π

(e) 1

9.

$$\int_0^1 \int_{2x}^2 4e^{y^2}\, dy\, dx =$$

(a) $e^4 + 1$

(b) $e^2 - 1$

(c) e^4

(d) $e^3 - 1$

(e) $e^4 - 1$

10. The graph of the function

$$f(x, y) = xy + \frac{1}{x} + \frac{1}{y}$$

has

(a) a saddle point at $(1, 1)$

(b) a relative minimum at $(1, 1)$

(c) a relative maximum at $(1, 1)$ and a relative minimum at $(-1, -1)$

(d) no relative extreme values

(e) a relative maximum at $(1, 1)$

11. The volume of the solid inside both $x^2 + y^2 + z^2 = 4$ and $(x - 1)^2 + y^2 = 1$ is equal to

(a) $\frac{32}{9}(\pi - 1)$

(b) $\frac{16}{9}(3\pi - 4)$

(c) $\frac{16}{9}(\pi - 4)$

(d) $\frac{4}{9}(3\pi - 4)$

(e) $\frac{16}{3}(\pi - 2)$

12.

$$\int_1^4 \int_0^1 \int_0^x 2z\, e^{-x^2}\, dy\, dx\, dz =$$

(a) $\frac{15}{2}\left(1 - e^{-1}\right)$

(b) $8\left(1 - e^{-1}\right)$

(c) $\frac{7}{2}\left(1 - e^{-1}\right)$

(d) $15\left(e^{-1} - 1\right)$

(e) $\frac{15}{2}\left(1 + e^{-1}\right)$

13. The volume of the solid region bounded above by the paraboloid $z = 4 - x^2 - y^2$ and the plane $z = 4 - 2x$ is equal to

(a) $\int_{-2}^{2} \int_{0}^{\sqrt{2x-x^2}} (2x - x^2 - y^2)\, dy\, dx$

(b) $\int_{0}^{\sqrt{2}} \int_{-\sqrt{2}}^{\sqrt{2}} (2x - x^2 - y^2)\, dy\, dx$

(c) $\int_{0}^{2} \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} (2x - x^2 - y^2)\, dy\, dx$

(d) $\int_{-4}^{4} \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} (2x - x^2 + y^2)\, dy\, dx$

(e) $\int_{0}^{4} \int_{-\sqrt{2x-x^2}}^{\sqrt{2x-x^2}} (4 - x^2 - y^2)\, dy\, dx$

14. The vector component of $\vec{u} = \langle 8, 2, 0 \rangle$ orthogonal to $\vec{v} = \langle 2, 1, -1 \rangle$ is

(a) $\langle 6, 3, -3 \rangle$

(b) $\langle -2, 1, -3 \rangle$

(c) $\langle 4, 1, 1 \rangle$

(d) $\langle 8, 2, 0 \rangle$

(e) $\langle 2, -1, 3 \rangle$

15. The area of the parallelogram that has the vectors $\vec{u} = \langle 1, 2, 2 \rangle$ and $\vec{v} = \langle 2, 1, -2 \rangle$ as adjacent sides is equal to

(a) 6

(b) 10

(c) 4

(d) 8

(e) 9

16.

$$\lim_{(x,y)\to(0,0)} \frac{\sin(x^2+y^2)}{x^2+y^2} =$$

(a) π

(b) -1

(c) 1

(d) 0

(e) $-\pi$

17.

$$\int_1^e \int_0^y \frac{4}{x^2+y^2}\,dx\,dy =$$

(a) 4π

(b) $\dfrac{\pi}{2}$

(c) 3π

(d) π

(e) 2π

18. The volume of the solid bounded by $z = 8(1-xy)$, $z = 0$, $y = x$, $y = 1$ and $x = 0$ is equal to

(a) 3

(b) 7

(c) 2

(d) 5

(e) 1

19.

$$\int_0^2 \int_0^x \sqrt{x^2+y^2}\,dy\,dx + \int_2^{2\sqrt{2}} \int_0^{\sqrt{8-x^2}} \sqrt{x^2+y^2}\,dy\,dx =$$

(a) $\dfrac{2\sqrt{2}\,\pi}{3}$

(b) $\dfrac{\sqrt{2}\,\pi}{3}$

(c) $4\sqrt{2}\,\pi$

(d) $\dfrac{8\sqrt{2}\,\pi}{3}$

(e) $\dfrac{4\sqrt{2}\,\pi}{3}$

20. The domain of the function $f(x,y) = \dfrac{\sqrt{x^2+y^2-9}}{e^{x+y}}$ is

(a) $\{(x,y):\ x^2+y^2>9,\ y=0\}$

(b) $\{(x,y):\ x^2+y^2\leq 9\}$

(c) $\{(x,y):\ x^2+y^2\geq 9,\ x+y\neq 0\}$

(d) $\{(x,y):\ x^2+y^2\geq 9,\ x=0\}$

(e) $\{(x,y):\ x^2+y^2\geq 9\}$

Answer Key

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Q	Ans	Brief explanation
1	**(e)**	$\frac{dy}{dx} = \frac{\sin\theta}{1-\cos\theta}$; at $\frac{\pi}{6}$: $\frac{1}{2-\sqrt{3}} = 2+\sqrt{3}$.
2	**(e)**	Interior value 0; edges give max 6 at $(\pm 2, \pm 1)$, min $-\frac{1}{4}$: $m + M = \frac{23}{4}$.
3	**(d)**	Spherical: $\rho \le 2$, $\frac{\pi}{4} \le \varphi \le \frac{\pi}{2}$: $2\pi \cdot \frac{8}{3} \cdot \frac{\sqrt{2}}{2} = \frac{8\sqrt{2}\pi}{3}$.
4	**(c)**	$\nabla f = \frac{1}{\sqrt{21}}\langle 1, 4, 2\rangle$ has length 1, so the maximum is 1.
5	**(c)**	$\nabla F \parallel \langle -4, 8, -2\rangle$: $(a, b, c) = (-2k, k, -k)$, $k = -1$: $P = (2, -1, 1)$, sum 2.
6	**(a)**	$V = xy\frac{30-5x-3y}{2}$; critical point $(2, \frac{10}{3}, 5)$ gives $V = \frac{100}{3}$.
7	**(e)**	Lagrange: $y = \frac{3x}{4}$, $x^3 = 8$, $(2, \frac{3}{2})$, $f = \frac{11}{2}$.
8	**(a)**	$\iint \sin(x+y)\, dA = \int_0^{\pi} 2\cos y\, dy = 0$, so the average is 0.
9	**(e)**	Swap order: $\int_0^2 \int_0^{y/2} 4e^{y^2}\, dx\, dy = \int_0^2 2ye^{y^2} dy = e^4 - 1$.
10	**(b)**	Only critical point $(1, 1)$; $d = 2 \cdot 2 - 1 = 3 > 0$, $f_{xx} = 2 > 0$: relative minimum.
11	**(b)**	Cylindrical: $V = 4\int_0^{\pi/2} \int_0^{2\cos\theta} r\sqrt{4 - r^2}\, dr\, d\theta = \frac{16}{9}(3\pi - 4)$.
12	**(a)**	$\int_0^x \to 2zxe^{-x^2}$, $\int_0^1 \to z(1 - e^{-1})$, $\int_1^4 z\, dz = \frac{15}{2}$.
13	**(c)**	Base disc $(x-1)^2 + y^2 \le 1$, height $2x - x^2 - y^2$, $x \in [0, 2]$.
14	**(e)**	proj $= 3\langle 2, 1, -1\rangle = \langle 6, 3, -3\rangle$; $\vec{u} -$ proj $= \langle 2, -1, 3\rangle$.
15	**(e)**	$\vec{u} \times \vec{v} = \langle -6, 6, -3\rangle$, length $\sqrt{81} = 9$.
16	**(c)**	With $u = x^2 + y^2 \to 0$: $\lim \frac{\sin u}{u} = 1$.
17	**(d)**	Inner: $\frac{4}{y} \arctan \frac{x}{y}\big|_0^y = \frac{\pi}{y}$; then $\int_1^e \frac{\pi}{y} dy = \pi$.
18	**(a)**	$V = \int_0^1 \int_0^y 8(1 - xy)\, dx\, dy = 8\left(\frac{1}{2} - \frac{1}{8}\right) = 3$.
19	**(e)**	Union is the sector $0 \le \theta \le \frac{\pi}{4}$, $r \le 2\sqrt{2}$: $\frac{\pi}{4} \cdot \frac{(2\sqrt{2})^3}{3} = \frac{4\sqrt{2}\pi}{3}$.
20	**(e)**	Only the root restricts: $x^2 + y^2 \ge 9$; e^{x+y} is never 0.